UNITED STATES OF AMERICA
BEFORE THE FEDERAL ENERGY REGULATORY COMMISSION

Fast- Tracking Power for American AI, Keeping Rates Low, and Modernizing the Grid	**)))**	**Docket No. RM26-4-000**

COMMENTS OF AMERICAN TERAWATT, INC

American Terawatt, Inc. ("American Terawatt") submits these opening comments in response to the Secretary of Energy's letter on the interconnection of large electrical loads and the associated Advanced Notice of Proposed Rulemaking ("DOE ANOPR Proposal"), as invited by the Commission's notice issued on October 27, 2025, in the above-referenced docket.1 The matters raised in this proceeding are highly consequential, with direct implications for the Commission's obligations under the Federal Power Act and for the nation's broader economic and security interests. In light of these considerations, American Terawatt urges the Commission to focus swiftly and pragmatically on reforms that can be adopted without delay and that will encourage the timely expansion of digital and energy infrastructure. Such reforms should be designed to make the most effective use of existing transmission and distribution assets while avoiding unnecessary burdens on the electric grid.

I. Introduction

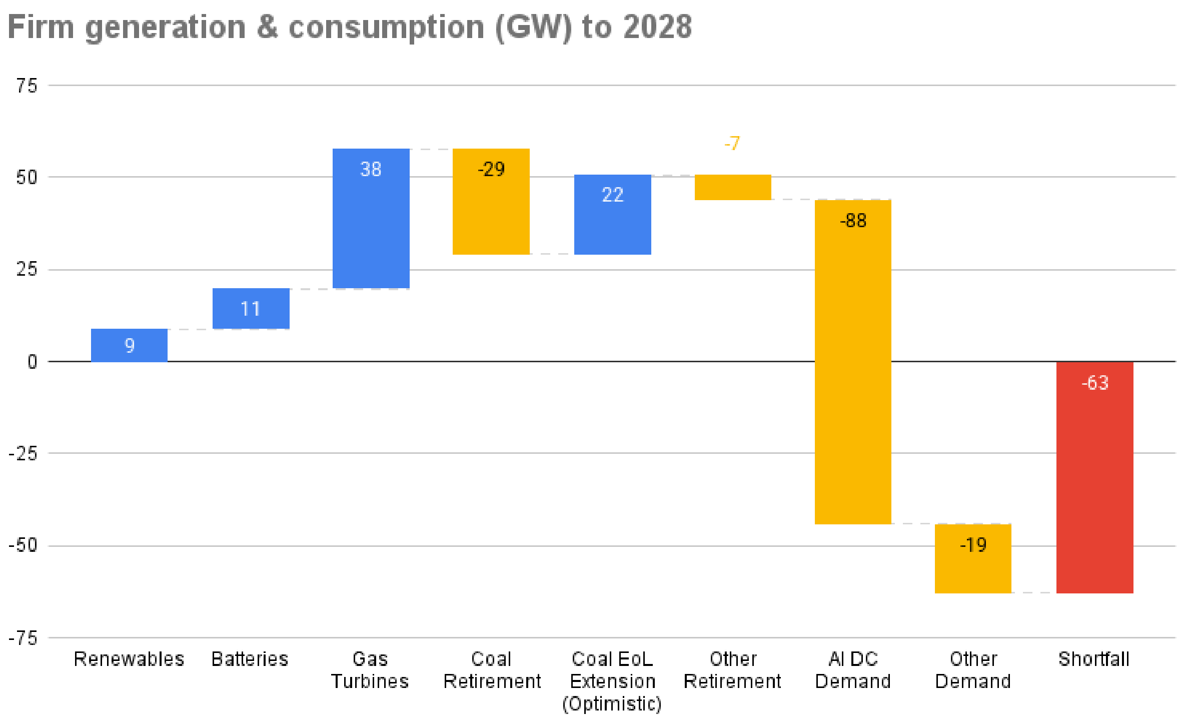
On October 23, 2025, the secretary of energy Chris Wright, in recognition of the crisis faced by the US electric grid, issued an advance notice of rulemaking to the Federal Electricity Regulatory Authority (FERC):

"To usher in a new era of American prosperity, we must ensure all Americans and domestic industries have access to affordable, reliable, and secure electricity. To do this, large loads, including AI data centers, served by public utilities must be able to connect to the transmission system in a timely, orderly, and non-discriminatory manner. This is an urgent issue that requires prompt attention."

This position paper explains how targeted technical and policy interventions can unlock the electricity needed to power American AI while keeping rates low and modernizing the grid by correcting the misaligned incentives between the operators of the American electrical grid, and data-center developers.

II. Problem

Growth in electricity demand from AI data centers is outpacing the capacity of the U.S. electric grid. By some estimates, electricity generating capacity will fall short of projected demand by double-digit gigawatts (GW) through at least 2028, and possibly beyond even if the planned retirement of coal-fired power plants is delayed or halted.



Based on predicted AI revenue. Source: SemiAnalysis

Even when power is available, actually getting data centers connected to the grid takes years. Traditional interconnection of large loads averages five years, and often up to eight years throughout the country. Our ability to bring compute for AI online is severely constrained by the availability of power.

If it continues to take this long to bring new compute capacity online, the United States will lose its competitive edge in AI. The competition is stiff: China's ability to rapidly build power generation and transmission infrastructure allows it to offset the inefficiency of its domestic AI chips—it doesn't matter if China's GPUs are four times less efficient[1] if it can build four times as many power plants as the United States has.

To secure power more quickly, data-center developers are increasingly resorting to expensive, risky solutions such as building and operating their own behind-the-meter natural gas power plants colocated with large datacenter campuses.[2]

This strategy has clear limits. It's hard enough to build and run a gigawatt-scale data center, let alone one with its own attached gigawatt-scale power plant. Instead of distributing redundancy, risk, and capital expenditure across many generators and loads, off-grid generation plants like these concentrate it at each new site. As data-center developers themselves have come to recognize, this is untenable in the long term.

How did we get here? Why can't the American electric grid bring the necessary gigawatts online fast enough to remain ahead of the United States's adversaries in AI?

The core issue is not capital. The real constraint is incentives. Utilities operate under cost-of-service regulation, where capital expenditures must be approved by regulators and then added to "rate base" to earn a regulated return.

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[1] Dylan Patel et al., "Huawei AI CloudMatrix 384—China's Answer to Nvidia GB200 NVL72," *SemiAnalysis*, April 15, 2025, https://newsletter.semianalysis.com/p/huawei-ai-cloudmatrix-384-chinas-answer-to-nvidia-gb200-nvl72.

[2] "GE Vernova and Crusoe Announce Major 29-Unit Aeroderivative Gas Turbine Deal to Deliver Power to AI Data Centers," Crusoe, July 22, 2025, https://www.crusoe.ai/resources/newsroom/ge-vernova-and-crusoe-announce-major-29-unit-gas-turbine-deal; Jeremie Eliahou Ontiveros et al., "xAI's Colossus 2—First Gigawatt Datacenter in the World, Unique RL Methodology, Capital Raise," *SemiAnalysis*, September 16, 2025, https://newsletter.semianalysis.com/p/xais-colossus-2-first-gigawatt-datacenter.

This structure rewards utilities for undertaking larger, slower, and more capital-intensive projects, because those projects expand rate base, even when smaller or faster solutions would deliver usable grid capacity sooner.

At the same time, many upgrade costs are ultimately recovered through rates paid by existing customers. Ratepayers are highly sensitive[3] to near-term increases in their bills, even when improvements would lower system costs over time. Regulators and elected officials respond to that pressure, which pushes planning toward minimizing visible, short-term rate impacts rather than optimizing for long-term efficiency and speed.

Grid operators and legislatures have begun to respond to the rapidly increasing demand for power by imposing specialized, data-center-specific power tariffs,[4] and mandating that new data-center loads must be curtailable during grid peak times.[5]

This solves part of the problem. Because the grid is built for peak load, new loads that can reduce load (curtail) during peak times considerably reduce the need for expensive grid upgrades, keeping rates stable for other customers.

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[3] Marc Levy, "As Electric Bills Rise, Evidence Mounts That Data Centers Share Blame. States Feel Pressure to Act," *Associated Press*, August 8, 2025, https://apnews.com/article/electricity-prices-data-centers-artificial-intelligence-fbf213a915fb574a4f3e5baaa7041c3a.

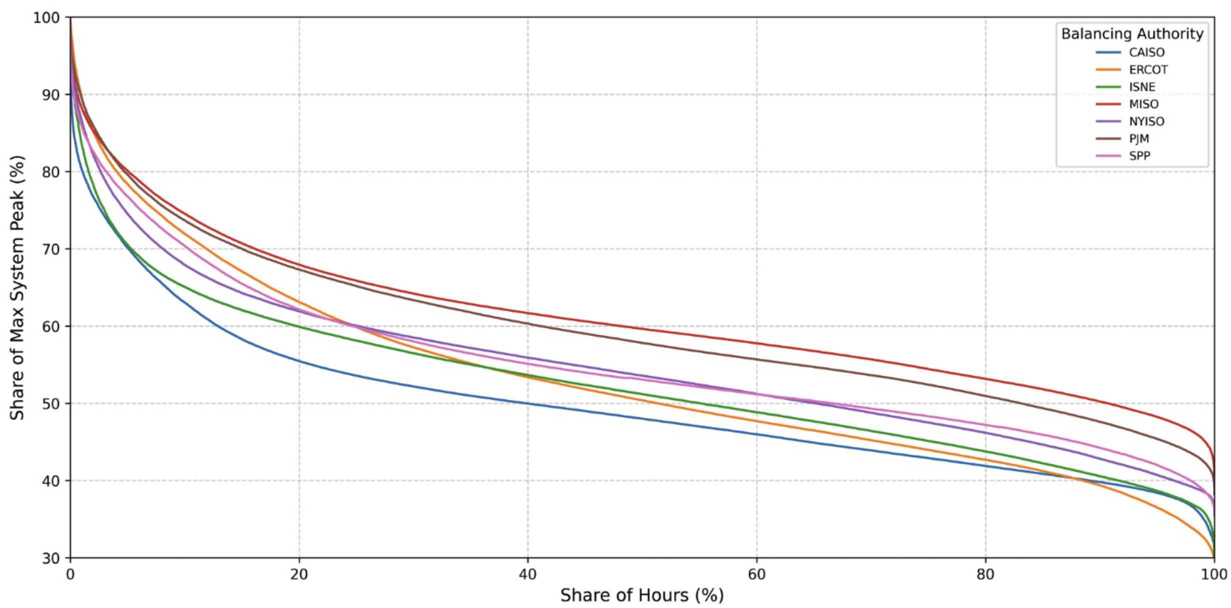
[4] Electricity: Data Centers: Rate Schedule and Requirements, H.B. 900, Maryland General Assembly (2025), https://mgaleg.maryland.gov/2025RS/fnotes/bil_0000/hb0900.pdf; Relating to Large Energy Use Facilities; and Declaring an Emergency, H.B. 3546, 83rd Oregon Legislative Assembly (2025), https://olis.oregonlegislature.gov/liz/2025R1/Downloads/MeasureDocument/HB3546/Enrolled.

[5] Relating to the Planning for, Interconnection and Operation of, and Costs Related to Providing Service for Certain Electrical Loads and to the Generation of Electric Power by a Water Supply or Sewer Service Corporation, S. 6, 89th Texas State Legislature (2025–2026), https://legiscan.com/TX/text/SB6/id/3248460; Stu Bresler and Tim Horger, "Large Load Additions PJM Conceptual Proposal and Request for Member Feedback," PJM, August 18, 2025, https://www.pjm.com/-/media/DotCom/committees-groups/cifp-lla/2025/20250818/20250818-item-03---pjm-conceptual-proposal-and-request-for-member-feedback---presentation.pdf.

Industrial load flexibility is not new. Many large electricity consumers voluntarily enroll in "demand response" programs,[6] which provide incentives to loads that curtail their operation during peak times. The bitcoin mining industry, where margins depend directly on electricity prices, has also long practiced "economic curtailment," reducing usage when prices spike.

Roughly half of the United States' electric infrastructure is idle at any given time. Existing off-peak grid capacity could accommodate an *additional 100 GW* of flexible loads, more than enough to cover data-center growth and other demand through 2028 and beyond, without needing new power plants.[7]

Load Duration Curve for US RTO/ISOs, 2016–2024



[6] "Demand Response," Electric Reliability Council of Texas, accessed October 25, 2025, https://www.ercot.com/services/programs/load.

[7] Tyler H. Norris, Tim Profeta, Dalia Patino-Echeverri, and Adam Cowie-Haskell, *Rethinking Load Growth: Assessing the Potential for Integration of Large Flexible Loads in US Power Systems*, Nicholas Institute for Energy, Environment, and Sustainability, February 2025, https://dukespace.lib.duke.edu/items/bb350296-d7a1-4d8f-acb0-2fba9b1f03de.

Source: Norris et al., “Rethinking Load Growth.”

However, flexibility mandates have received significant pushback from the AI data-center industry.[8] Alone, they do nothing to reduce interconnection delays, and because of the economics of AI compute and uptime service-level agreements (SLAs), which require upwards of 99.999 percent availability as demanded by customers, curtailment strategies that shut down GPUs are nonstarters.

Moreover, flexibility mandates and differential tariffs are being implemented haphazardly by utilities, grid operators, and state legislatures, creating a regulatory snarl that threatens to further delay data-center projects in the United States.

III. Solutions

These problems can be solved in a way that benefits all parties. The proposed solutions come in two forms: technical measures that allow for AI data-center load flexibility and policy that aligns incentives between grid operators and data-center developers.

A. Technology: Grid Responsive AI Datacenters (GRAIDs)

Uptime SLAs are critical to compute customers such as frontier AI labs because checkpointing and restarting large-scale AI workloads such as training runs introduces significant engineering complexity, increasing costs and reducing reliability. However, by throttling GPU power and slowing computations rather than switching devices off entirely,

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[8] *Critical Issue Fast Path—Large Load Additions Stakeholder Comments*, PJM, August 28, 2025, https://www.pjm.com/-/media/DotCom/committees-groups/cifp-lla/postings/20250828-stakeholder-comments-cifp-lla.pdf.

grid-responsive AI data centers (GRAIDs) can maintain uptime SLAs without ever stopping workloads.

Modern GPUs have a wide operating power envelope, allowing for a large degree of throttling. For example, the Nvidia H100 SXM5 has a peak power consumption of 700 watts (W) but can be throttled to 200 W on demand—a reduction of 71 percent—without interrupting computation.

In this way, GRAIDs can respond to peak grid load by throttling GPUs as required to reduce power while maintaining uptime in response to demand response (DR) signals from the grid operator.

Long-running workloads such as large-scale pretraining would continue uninterrupted. Time-insensitive inference jobs may also be throttled, and latency-critical jobs could either be routed to unthrottled facilities or assigned to a pool of "hot," unthrottled compute kept in reserve.

Here's how it would work in practice at the level of a data center:

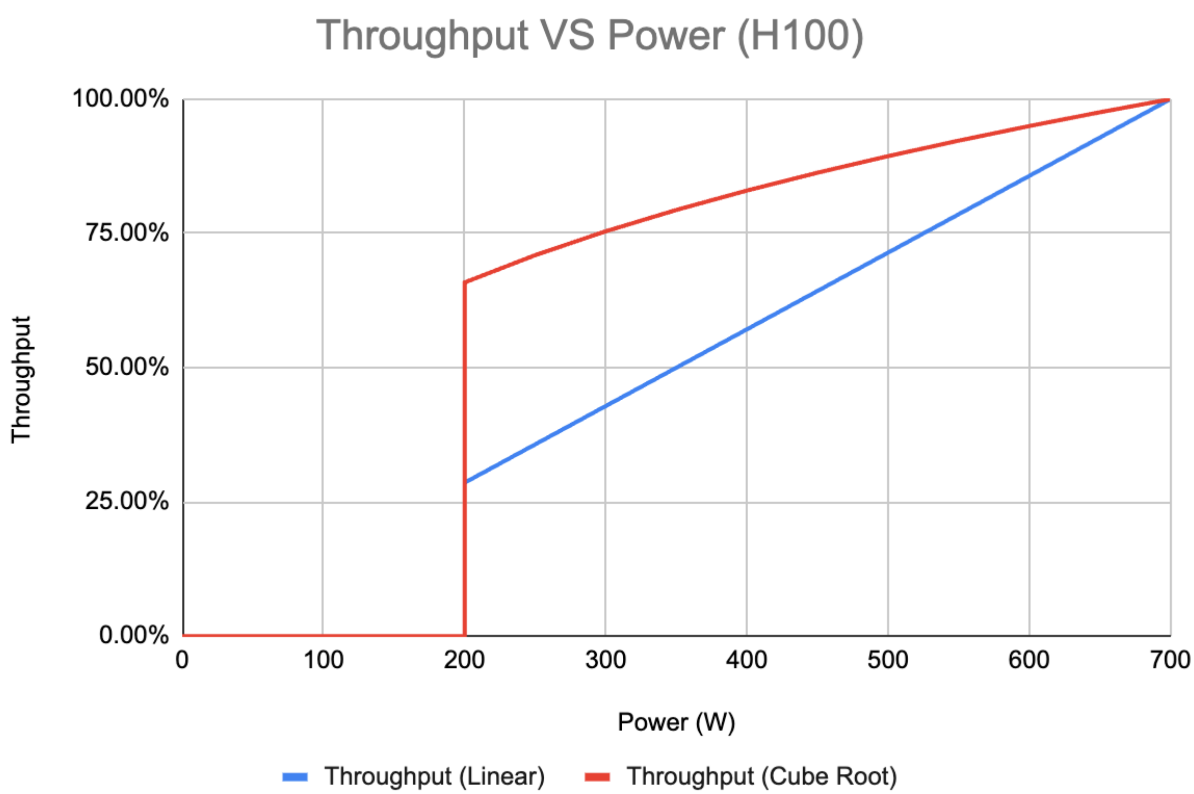
- The data-center operator would coordinate with the utility to curtail an agreed-upon fraction of load on demand in response to grid conditions as part of a DR program.
- An automated software controller at the data center would monitor for DR events from the grid operator. This could be accomplished using the open protocol OpenADR[9] over an HTTPS connection, with redundant backups.

[9] "Connecting Smart Energy to the Grid," OpenADR Alliance, accessed October 25, 2025, https://www.openadr.org/.

- After the signaling of a DR event, the data-center power controller would automatically throttle GPUs according to the agreed-upon policy by setting power caps at the level of the baseboard management controller (BMC) (e.g., via Redfish).[10]

This method would integrate seamlessly with current infrastructure and would not require any new hardware or software from compute customers. Since power throttling would be executed at the BMC, data-center operators would never access the host environment and would not be able to view customer workloads or data.

The trade-off would be in throughput. Computations would be slowed as power was reduced, with the degree of slowdown depending on the specific device and workload.



[10] "Redfish APIs Support," Nvidia, last updated September 10, 2025, https://docs.nvidia.com/dgx/dgxb200-user-guide/redfish-api-supp.html#power-capping.

Approximate throughput envelope, H100 SXM5. Real throughput lies between the linear and cube root bounds.

Throttling is nonlinear with respect to power, partly because semiconductors are themselves nonlinear devices and partly because GPUs have other power-consuming components beyond the core chip. In theory, the true degree of throttling lies between a best-case cube-root curve and a worst-case linear curve. In practice, the throughput curve can be measured empirically per device and workload.

Because grid peak events are rare and typically last about an hour, the impact to overall throughput is small even under worst-case throttling. Consider the ERCOT grid in Texas, where emergency response curtailment can occur for up to 60 hours per year and where Four Coincident Peak (4CP) or near-CP events occurred for 41 hours in 2024.[11]

This number represents at worst an annual throughput reduction of *only 1.1 percent*, or, in other words, access to 98.9 percent more capacity than if those chips sat idle awaiting grid upgrades and interconnection.

Including facility overhead, a 100-megawatt (MW) GRAIDs facility could curtail approximately 50 MW on demand, all while keeping workloads continuously running. For compute customers, the marginal loss in throughput would be tolerable, especially when compared to the substantial increase in available capacity that this approach would make possible. A typical 100-day training run would take one day longer, but many more such runs could take place.

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[11] *2024 Annual Report on Demand Response in the ERCOT Region*, Electric Reliability Council of Texas, updated February 2025, https://www.ercot.com/misdownload/servlets/mirDownload?doclookupId=1080511461.

B. Policy: Aligning Incentives at the Federal Level

Through their behind-the-meter and off-grid projects, data-center developers have demonstrated their willingness to pay for faster time-to-power. Simultaneously, utilities and grid operators have recognized the virtues of flexible loads as a way to keep rates low and acknowledge that revenue from these loads can be used to modernize the grid. The right policy intervention would align these interests.

Data-center developers willing to build flexible AI factories or retrofit existing sites with flexible load capabilities should have access to a significantly accelerated grid interconnection process. By charging a predetermined up-front premium for this form of accelerated interconnection, utilities need not incur increased rates for consumers in the short term.

In the long term, thanks to significant revenue from new large flexible loads and capped returns for utilities, electricity costs might actually be *reduced* for other ratepayers.

Looking to the future, a reindustrializing America will need power for other large loads. Industrial projects such as chip fabs, rare earth mineral refineries, and highly automated factories draw hundreds of megawatts and may not have the flexibility of AI data centers. Revenues from flexible data-center loads that are interconnected today will pay for the grid upgrades necessary for the loads of tomorrow, again without raising rates for ordinary Americans.

The Federal Energy Regulatory Commission (FERC) has so far explicitly chosen not to regulate flexible loads, leaving them to individual grid operators. It's time for that to change.

We propose the following:

- FERC should direct independent system operators (ISOs) to develop expedited interconnection processes for defined large flexible loads, with transparent criteria. FERC should direct regional transmission organizations and ISOs to propose enabling, nondiscriminatory tariffs for participation in expedited interconnection processes for defined large flexible loads.
- To ensure clarity for both developers and grid operators and prevent a national patchwork of regulations, FERC should define qualifying flexible loads for the purposes of transmission-level interconnection and wholesale market participation.

Regulatory action of this magnitude from FERC, targeting specific types of loads and grid operations, is not unprecedented. In 2011, FERC issued Order 1000, which required transmission providers to participate in regional transmission planning processes, not just local or utility-specific planning. Similarly, in 2018, FERC issued Order 841, requiring grid operators to open their wholesale markets to energy storage. FERC Order 2222, issued in 2021, ensured aggregators of distributed energy resources could participate in wholesale energy, capacity, and ancillary markets such as DR programs.

And, perhaps most relevant, FERC Order 2023, issued in 2023, directed grid operators to simplify interconnection procedures for new generating assets. Now, FERC can do the same for loads.

Indeed, Secretary Wright's directive includes directives related to expedited interconnection for curtailable large loads (section 24), and transmission network upgrades paid for by facilities developers (section 25), are great steps toward these goals.

We believe these policy interventions will not only rapidly expand the availability of power for American AI but keep rates low for consumers and rapidly modernize the whole electric grid for future American industrial load.

Respectfully Submitted,

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